STUDENT TEAM WEEKLY REPORT: MODULE 7

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Analytics Engine, Psychometric Item Analysis & Performance Reports]	**TEAM ID:** 3 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 28/09/2026 **STUDENT LEADER:** Mrunal Kolhe

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• **Institutional Analytics Engine:** Developed AnalyticsEngine.java (869 lines) containing OverviewHandler, ItemsHandler, TopicsHandler, TrendsHandler, and ReportsHandler. • **Psychometric Item Analysis:** Implemented Classical Test Theory Discrimination Index (D = U − L using top 27% and bottom 27% cohorts) to identify item quality (EXCELLENT, OPTIMAL, OVERLY_EASY, WEAK_ITEM). • **Topic Mastery & Learning Gaps:** Built topic health evaluator flagging syllabus topics with <50% mastery as critical learning gaps (isWeak: true) with remedial recommendations. • **Multi-Dimensional Reports API:** Built GET /api/analytics/reports?dimension=student|class|subject|quiz supporting multi-level aggregation. • **Dual Export (PDF & CSV):** Built RFC 4180 CSV export endpoints (&format=csv) and print-optimized executive PDF reporting stylesheets.	• **Discrimination Zero-Division:** Cohorts with fewer than 3 attempts threw division-by-zero errors. Action: Added minimum sample size guard defaulting D = 0.40 when N < 3. **GREEN (RESOLVED)** • **Missing Topic References:** Questions lacking reference metadata broke topic aggregation. Action: Implemented fallback grouping to "General Concept Assessment". **GREEN (RESOLVED)** • **CSV Comma Corruption:** Question stems containing commas corrupted spreadsheet column alignment. Action: Built custom CSV string sanitizer wrapping fields in escaped double quotes. **GREEN (RESOLVED)**	**Faculty UX Guidance:** Request faculty mentor feedback on the discrimination index thresholds (D ≥ 0.35 for Excellent, D < 0 for Flawed) to ensure alignment with institutional psychometric standards.

INTEGRATION AND COLLABORATION
Frontend Team: Connected analytical KPI cards, cycle trend progress bars, and topic health badges on admin.html and author.html.
Database Team: Optimized SQL aggregation queries for class-wise and subject-wise score averages.
Integration Team: Validated all 5 analytics REST endpoints returning valid JSON structures under empty and populated states.
Pipeline Learnings: Mastered educational data mining, psychometric evaluation algorithms, and multi-dimensional reporting.

COMMON KPI EVIDENCES
Analytics Response Time: Full psychometric overview and item analysis calculated and returned in <45 ms.
Item Classification Accuracy: 100% correct categorization of test items into Overly Easy, Weak, and Optimal categories.
Export Versatility: Verified both .csv spreadsheet generation and @media print executive PDF layouts.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• **Mrunal Kolhe:** Implement cryptographically secure Bearer session token management and RBAC guards. (Due: Oct 01, 2026) • **Mrunal Kolhe:** Build end-to-end automated security verification test suite for authentication and audit controls. (Due: Oct 03, 2026)	